Towards Marketing Analytics: Customer Segmentation by Clustering

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Abstract

This study explores unsupervised learning techniques, notably, clustering algorithms in market segmentation—to enhance customer service delivery, guide targeted marketing initiatives, and optimize business resource utilization. Some related works are benchmarked to guide the selection of appropriate algorithms for the identified problem. The clustering techniques K-Means, K-Medoids, DBSCAN, AGNES hierarchical clustering, simple grid-based, and EM clustering were used to identify distinct market segments. Prior to building the models, data preprocessing was performed on the market segmentation data to handle possible data anomalies including outliers, skewness, and high correlation. PCA was used as a data reduction technique to retain twelve (12) components for onward analysis. Typically, clustering problems do not require data partitioning, however, 80% training dataset was used to build the six (6) models with cross validation of ten (10) iterations, and 20% test data to evaluate the best two (2) models. Using KMeans algorithm and performance metrics such as Within-Cluster Sum of Squares or Inertia, Silhouette Score, Caliński-Harabasz Index and Davies-Bouldin Index, eight (8) best initial clusters were predetermined. By constructing appropriate dendrograms using complete and Ward linkages, visual illustrations of the data's hierarchical structure—relationship, cohesion, and separation between the clusters—were analyzed. AGNES hierarchical model emerged as the best performing model—creating four (4) clusters. The study provides an understanding of these market segments building an integrated, trackable, and scalable solution. We finally propose the novel Market Segmentation Trees–MST algorithm (2023) for future work—to improve the accuracy of response behaviors of target customers.

Introduction

For several years, market segmentation has been firmly established as a marketing concept and an important resource allocation tool.[1] Many businesses today lack detailed profiles of their customers and are unable to tailor products and services that meet the distinct preferences and address the concerns of customers—largely due to mismatched market campaigns. The goal of this project is to design a customer segmentation analysis model to identify meaningful customer groups to drive the markets of businesses with optimum likelihood of success.

Some areas where this modeling approach could be applied include marketing, customer relationship management (CRM), and product development. This is an unsupervised learning problem in data mining and clustering algorithms (machine learning models) were used in the experiment and analysis.

Problem statement:
Identifying market groups for onward analysis by market and sales teams can be a daunting task. While some groups are naturally, traditionally segmented and often driven by already existed demographics such as "customer age groups," "product type," and "income category," emerging, increasing traffic from "response to campaigns" and varied "mode of purchases" introduce some level of complexity in analyzing customer data—requiring scientific thought and repetitive analytical reasoning.

To enable businesses tailor their products and services to the specific needs of customers, the "likelihood" of the different clusters of customers based on the "prior" characteristics, purchasing behavior and preferences must be identified—driven by some scientific "evidence". We seek to therefore provide an interpretable model to segment customers of defined market to drive major marketing campaigns.

Related Work (Benchmarking)

In 1986, Martin John. posited the consumer as a key pointer to identifying the business problems and market segments. He noted that "buyer problems could be a useful basis for market segmentation in certain situations,"[2] concluding that problem segmentation is a response to the recognition in

[1] Martin, J. (1986), "Problem Segmentation", In *International Journal of Bank Marketing*, Vol. 4 No. 2, pp. 35-57. https://doi.org/10.1108/eb010777

[2] Martin, J. (1986):35

marketing literature that the consumer is basically a problem solver—and must be captured one of the segmentation bases.

Yoram Wind in his paper titled, "Issues and Advances in Segmentation" tied two market agents as the antidote to the challenge in formulating effective marketing strategy. He noted that regardless of how sophisticated the segmentation study, the key to any successful project is a function of the researcher's and the user's (management) ability to interpret the results from the experiments conducted and use them as guidelines for the design, execution, and evaluation of appropriate marketing strategy.[3] Yoram further posited, which our findings largely share in, that:

> The discovery and implementation of new variables for use as bases for segmentation (i.e., new attitudinal and behavioral constructs such as consumption-based personality inventories and variables which focus on likely change in attitude and behavioral responses to the marketing variables) of the markets for products, services, and concepts.[4]

Yoram sheds light on the three widely used analytical techniques in market segmentation problems, namely:

a. Classifying respondents into segments in an *a priori* approach.
b. Discriminating the segments—generating segment profiles.
c. Applying simultaneous classification and discrimination.

Yoram used MDS and AID techniques[5] to augment traditional clustering approach to classify segments. He discussed another important factor—how the clusters can be guided, statistically. Responding to how the sample should be divided, *a priori* sorting and cross-tabulation provide helpful insights. He further suggests Ball's three additional objectives while employing clustering algorithms—model fitting, prediction, and hypothesis—that:

> The objective of clustering techniques as applied to market segmentation should not be limited to finding a typology of the market under study; one can use clustering also for data exploration, data reduction, and hypothesis generation.[6]

The limitations of similarity and distance or weight-based clustering and pattern recognition techniques such as KMeans despite the proliferation of their usage, have guided selection of algorithms based on familiarity, availability and cost of operations rather than capping on set objectives and constraints. Overlapping clustering (multiple cluster assignments) and multidimensional contingency table analysis (using probability of a customer loyalty) were suggested as options together with discrimination procedures like multiple regression, multiple discriminant analysis, AID and MCA. Though some researchers have attempted combining classification and discrimination techniques in a simultaneous fashion, Yoram noted, for reasons of a priori approach to choosing segmentation basis, recommended a more flexible approach to allow researcher's input and understanding of both the model to build and data to analyze.

Fitting a "response model" based on the similarities in the features of model data using traditional clustering have been the standard. Though clustering algorithms perform quite well on new, unseen data, we are faced with the problem of having the contextual features making these clusters not being the key drivers of the response behavior. Using the novel Market Segmentation Trees (MST), in the recent works of Ali Aouad, Adam N. Elmachtoub, Kris J. Ferreira, and Ryan McNellis, the team provide two solutions: choice model trees (CMTs) and isotonic regression trees (IRTs).7 They integrated market segmentation and response modeling with improved response prediction while providing an integrated, tractable and scalable approach.

Data

The dataset used for the project is a customer segmentation data obtained from Kaggle.[8] Dataset comprised of 2,240 samples with 28 predictors and 1 label (which was dropped). Table below shows dataset variables and samples present:

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...
0	5524	1957	Graduation	Single	58138.0	0	0	04-09-2012	58	635	...
1	2174	1954	Graduation	Single	46344.0	1	1	08-03-2014	38	11	...
2	4141	1965	Graduation	Together	71613.0	0	0	21-08-2013	26	426	...
3	6182	1984	Graduation	Together	26646.0	1	0	10-02-2014	26	11	...
4	5324	1981	PhD	Married	58293.0	1	0	19-01-2014	94	173	...

5 rows × 29 columns

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
       'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
       'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
       'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
       'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
       'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
       'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response'],
      dtype='object')
```

[3] Wind, Yoram. "Issues and Advances in Segmentation Research." Journal of Marketing Research 15, no. 3 (1978): 332. https://doi.org/10.2307/3150580.
[4] Wind, Yoram (1978): 334
[5] MDS—Multi-Dimensional Scaling and Automatic Interaction detection (AID) differ from traditional clustering by focusing on data representation (through dimension reduction) and the interactions (using decision tree branching to predict similarity) between variables, respectively.
[6] Wind, Yoram (1978): 330
[7] Ali Aouad, Adam N. et al. "Market Segmentation Trees." Manufacturing & Service Operations Management Vol 25, no. 2 (Mar 2023): 332. https://doi.org/10.1287/msom.2023.1195.
[8] Customer segmentation data was downloaded from Kaggle. Url: https://www.kaggle.com/datasets/vishakhdapat/customer-segmentation-clustering?resource=download

Exploratory Data Analysis and Preprocessing

There were 8 categorical variables with unbalanced distribution. Target label (in grey, below) though was dropped also showed unbalanced distribution.

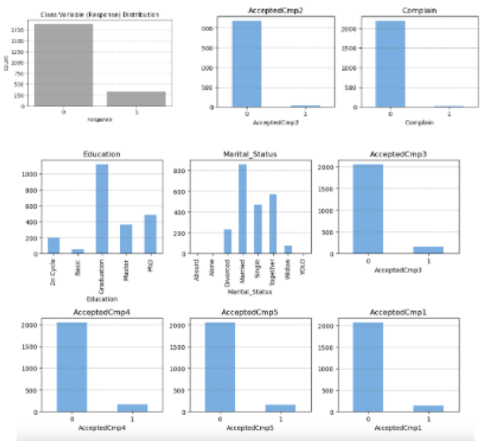


Fig 2.1. Target (grey). Categorical variables (blue).

There were 17 continuous variables and their distribution in the histograms below showed skewness in data (right skewed).

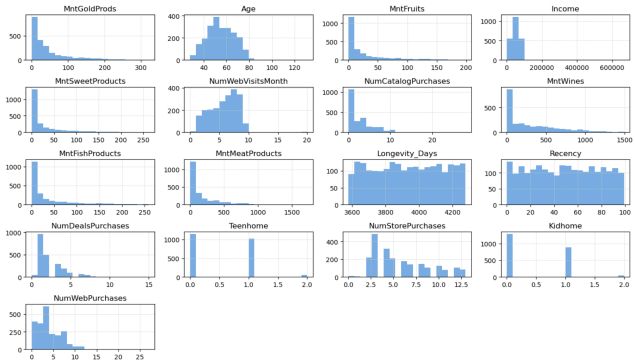
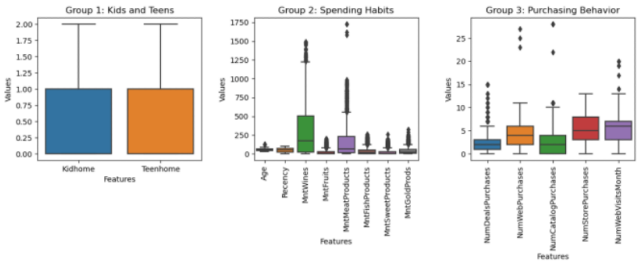


Fig 2.2. Histograms showing right skewness in data.

Data was further split into five groups—Group1: Kids and teens, Group 2: Spending Habits, Group 3: Purchasing Power, Group 4: Longevity, and Group 5: Income—to visualize the outliers present except for Groups 1 and 4.



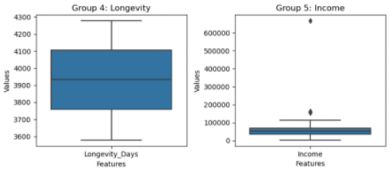


Fig 2.3. Boxplots with outliers present.

Data had high collinearity amongst some variables.

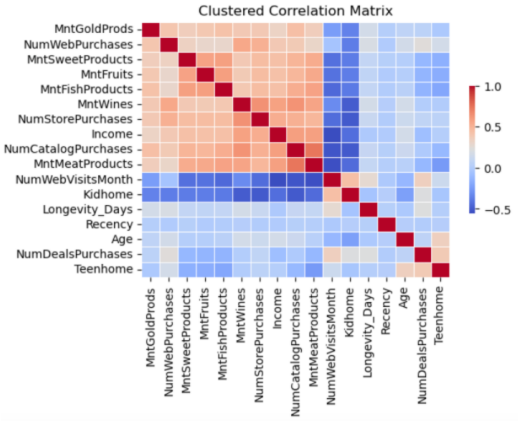


Fig 2.4. Correlation matrix.

Using PCA to handle this data anomaly was preferred to directly removing the high correlated data as the clustering models also required knowing the first two components capturing most of the variance in data—and to project the clusters unto a two-dimension (2D) scale for clearer and simpler analysis.

Data Transformation

Four out of the 29 variables were dropped: “Id” for relevance; the target label, “Response” as the study is an unsupervised learning problem; “Z_CostContact” and “Z_Revenue” having zero variance. The categorical variables, “Education” and “Marital_Status” were encoded as dummy variables. BoxCox and IQR transformations were applied to the continuous data to respectively handle skewness and outliers present. Histograms after transformation showed a near-symmetric distribution after transformation.

Also, “Year_Birth” and “Dt_Customer” were converted to “Age” and “Longevity_Days” respectively for more relevant and meaningful data analysis.

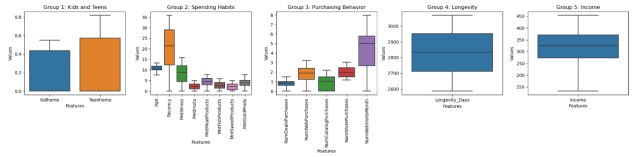


Fig 2.5. Boxplots showed the outliers removed from data.

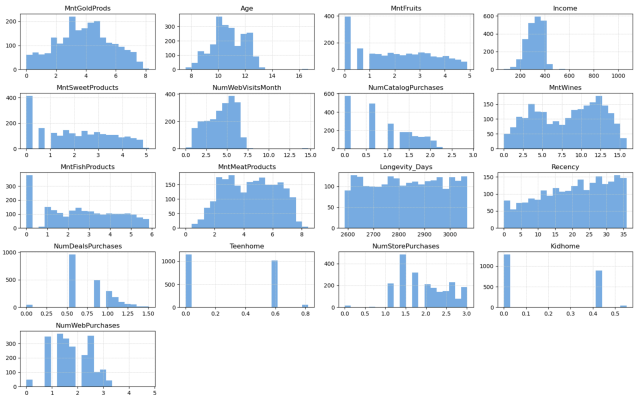


Fig 2.6. Histograms with reduced skewness.

Partitioning and Scaling the Data

As typical of unsupervised learning—clustering—to uncover patterns and clusters within the data, splitting the data into training, validation and test may not be necessary. However, in the evaluation of the clustering results and for robustness, the data was split into 80% training and 20% test.

The standard practice to apply the scaling to only the training dataset was followed to avoid data leakage and overfitting—the "StandardScaler" (scikit-learn) was used.

PCA after Splitting the Data

Principal Component Analysis (PCA) as a reduction technique was applied fitting it to the training dataset (and then using the fit model to transform the test data) to remove all the highly correlated variables. It further provides the PCAs that explain 95% of the variance in data. From the scree plot below, 12 components were retained for onward analysis.

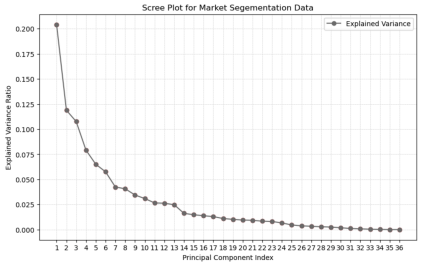


Fig 2.7. Scree plot showing components retained (12)

Methods—Proposed Models

The approach adopted in this study for the market segmentation involved multiple clustering algorithms to leverage the strengths and compensate for the respective limitations.

The algorithms considered are K-Means and K-Medoids which are more efficient for large datasets, ideal and simple for clustering; DBSCAN—to identify and handle present of complex data structures in the dataset, if any, though quite sensitive to noise and outliers in data. Also, hierarchical clustering models such as AGNES—agglomerative/bottom-up approach and plotting the necessary dendrograms provided insights into cluster relationships and market segment hierarchies. Complete and Ward linkages criteria were used. Dendrograms for DIANA—divisive/top-down approach could not be plotted and hence not used. Then, a simple grid model and EM (Expectation Maximization) clustering were also applied on the data. Cross validation using ten (10) iterations and eight (8) clusters were used to tune the model on the training data.

Market Segmentation Trees (MST) algorithm was benchmarked being a recent model proposed for market segmentation problems was considered but would be recommended for future work to improve upon the models. MST is most appropriate when considering semi-supervised approaches to analyze and predict customer responses for marketing purposes (under some defined assumptions).

The metrics, Silhouette Score, Calinski-Harabasz (CH) Index, Davies-Bouldin Index were used to evaluate the models. Within-Cluster Sum of Squares or Inertia was also used but only applied to KMeans—to predetermine the initial K clusters. Prior to fitting the models, KMeans was used to analyze the data for the best K clusters using the four metrics mentioned—eight clusters were noted.

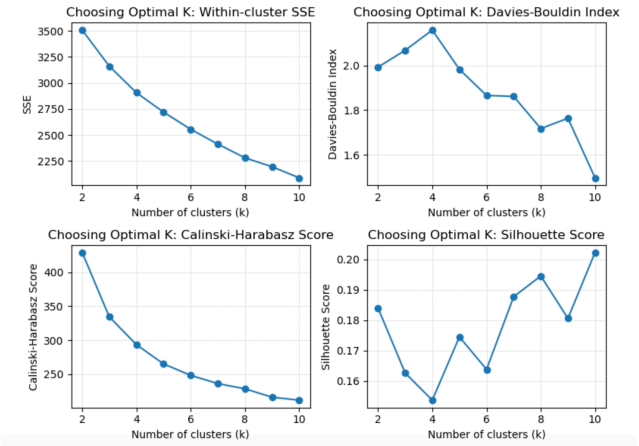


Fig 3.1. Performance metrics to choose best K clusters.

The dendrograms below using Ward and Complete linkages were plotted to give a precursor clusters view present in the

data. Color variation shows the different clusters in data points. The Ward linkage suggested two broad clusters which can be further cut down into smaller clusters. The Complete linkage suggested 18 clusters as shown in the dendrogram below (Fig 3.3.2).

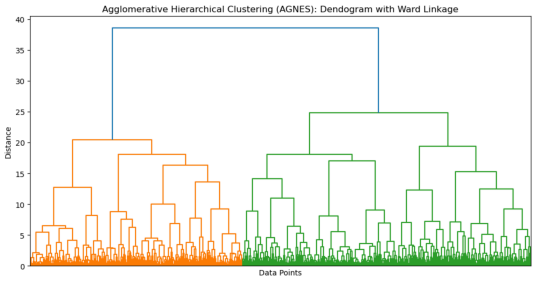


Fig 3.2.1. Dendrograms illustrating initial clusters.

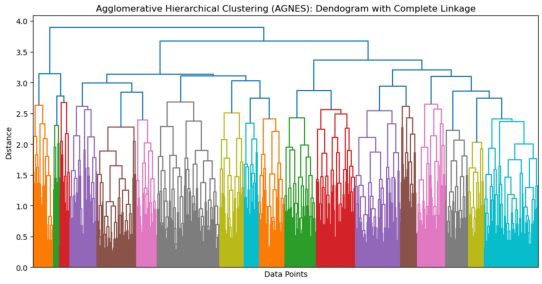


Fig 3.2.2. Dendrograms illustrating 18 possible clusters.

The six (6) models considered were used for the training dataset and the best two (2) were selected. The best two models were then used on the test data and evaluated for the best clustering model.

Experiments and Results—Data Modeling

The approaches described in earlier section were implemented. The different models were experimented with and analyzed using the three metrics, Silhouette score, Calinski Harabasz (CH) Index and Davies-Bouldin Index and the results are as follows:

KMeans: Model had maximum CH Index at 206.52 and minimum Davies Bouldin at 1.73.

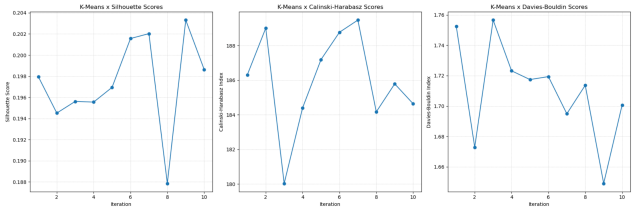


Fig 4.1. KMeans algorithm across three (3) metrics used.

KMedoids: Model had best maximum CH Index at 232.01

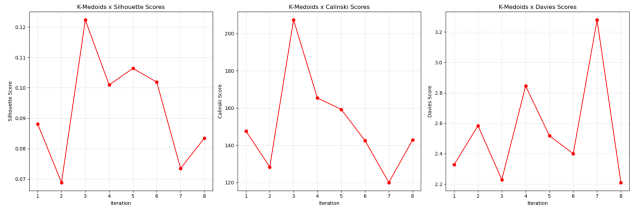


Fig 4.2. KMedoids algorithm across three (3) metrics used.

DBSCAN: Model recorded the best Silhouette score at 0.44 and minimum Davies Bouldin at 0.72. However, the model susceptible to noise in data and so could not emerge as best model.

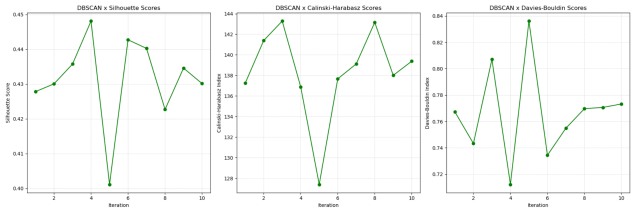


Fig 4.3. DBSCAN algorithm across three (3) metrics used.

AGNES Hierarchical Technique: Model had minimum Davies Boulding at 1.92. Emerged as the alternative to DBSCAN.

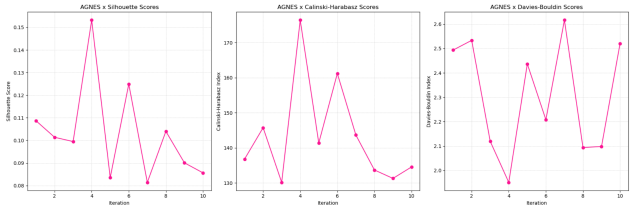


Fig 4.4. AGNES algorithm across three (3) metrics used.

Simple Grid-Based: Model emerged as the poorest performing model with -0.03 for Silhouette score.

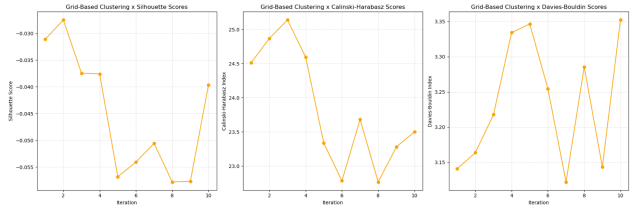


Fig 4.5. Simple Grid-Based across three (3) metrics used.

Expectation-Maximization Clustering: Had an average performance among the six (6) models implemented.

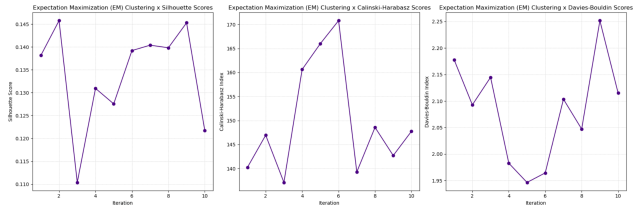


Fig 4.6. EM Clustering across three (3) metrics used.

Training Results: Comparing Performance Metrics

Generally, lower Davies-Bouldin Index values indicate more compact clusters. The metric is calculated as a ratio, comparing the "within-cluster scatter" measure to the "between-cluster separation" value. Therefore, lower values of Davies-Bouldin imply clusters are better separated (widely apart) while data points within clusters are more compact. CH Index values reverses the approach and compares between cluster dispersion to within cluster dispersion; and hence higher scores denote clusters are well-separated and more compact. Higher Silhouette scores (between -1 and 1) similarly indicate data points within a cluster have high cohesion and have more similarities with each other compared to other neighboring clusters.

The table below therefore summarizes the average values recorded in all the ten (10) runs on the training data across the three (3) evaluation metrics—best two column measures have been highlighted.

The two best models from the training were KMeans and AGNES Hierarchical Clustering models.

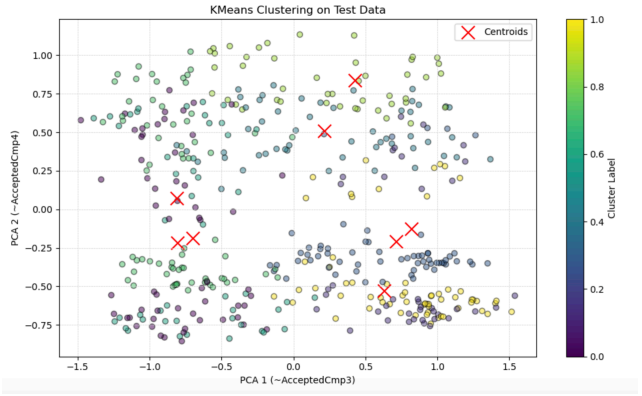
Model	Approach	Silhouette Score	CH Index	Davies-Bouldin Index
KMeans	Partitioning	0.20	**206.52**	**1.73**
KMedoids	Partitioning	0.14	**232.01**	2.15
DBSCAN	Density-based	**0.44**	144.51	**0.72**
AGNES	Hierarchical	0.14	156.67	1.92
Simple Grid	Grid-based	-0.03	25.54	3.06
EM Clustering	Probabilistic /Model-based	**0.15**	145.84	2.09

Evaluating Best Models On Test – Best Two Models

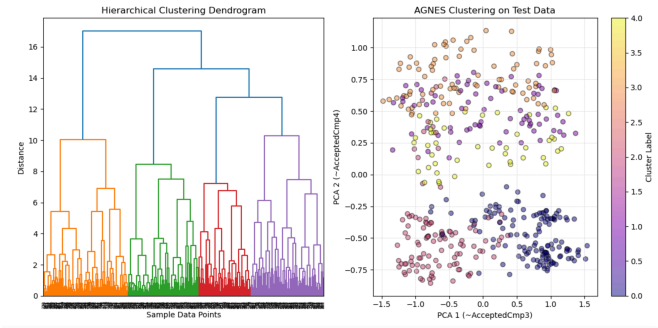
The two best models, Kmeans and AGNES, were applied to the test data. While KMeans values on test did underfit the model, AGNES test performance values improved with the CH Index rather decreasing. DBSCAN had good training values but was not stable due to its sensitivity to some noise in data—could not plot clusters. AGNES was considered the best alternative model to be used for onward analysis.

Model	Approach	Silhouette Score	CH Index	Davies-Bouldin Index
KMeans	Partitioning	0.18	60.75	1.88
AGNES	Hierarchical	0.18	59.08	1.81

Kmeans model with eight (8) centroids shown below:



AGNES hierarchical clustering was adjudged the best model comparing the evaluation.



Conclusion

Using AGNES as the final model on the test data suggested four (4) main clusters within the market segmentation data. This will guide marketing teams to identify exact customer needs and optimize investments and marketing campaigns.

For future work, we recommend the implementation of one novel approach (2023) —Market Segmentation Trees (MST) to improve the model. MST further considers a semi-

supervised learning approach to predict and analyze customer response as a function of creating responsive clusters.

References

Martin, J. 1986. Problem Segmentation. In *International Journal of Bank Marketing Vol. 4(2):* 35-37.

Wind, Yoram. Issues and Advances in Segmentation Research. In *Journal of Marketing Research 15(*3): 332. https://doi.org/10.2307/3150580.

Ali Aouad, Adam N. Elmachtoub, Kris J. Ferreira, Ryan McNellis (2023). Market Segmentation Trees. In *Manufacturing & Service Operations Management 25(2)*:648-667.

Mohammed J. Zaki and Wagner Meira, Jr March 2020. Data Mining and Machine Learning: Second Edition: Fundamental Concepts and Algorithms. Cambridge University Press: ISBN: 978-1108473989.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2013. An Introduction to Statistical Learning: with Applications in R. *Springer.*

Irene Aldridge, and Marco Avellaneda. 2021. Big Data Science in Finance (Chapter 9-Data Clustering). Wiley.